Slick Licks

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The Ambition

- Back in the late 1900s I was part of a jazz workshop helping people learn to play jazz.
- One of the exercises was called "slick licks" - we would sit in a circle and “trade licks” on our instruments.
- A big barrier for jazz learners can be finding people to play with.
- Q. Can I build a software system to enable people to practice their slick licks? And specifically… can a bass player use this?!
- It should listen and respond, sometimes lead, be realistic enough to be useful.
- NB This is like a highly interactive play-along for practice purposes, but it's not intended to replace a human player: human interaction was key to slick licks!

All the things I have been* thinking about

Phrase analysis

Question and answer

Leadership & role swapping

Contour steering

Chord conditioning

Harmonic modes

Scale pitch bias

Pitch range

Swing / triplet feel

Live tempo tracking

Dynamics

Articulation

Bass pitch-class tracking

Energy arc

Motivic development

Lyrical motif re-use

Voice leading

Modal leap bonus

Stochastic thinning

Phrase breathing

Singable duration bias

Rhythmic density

Rhythmic complementarity

Register contrast

Phrase length control

Beat-matching / trading bars

Ostinato

Proactive mode

Arc timing

Phrase-shape dynamics

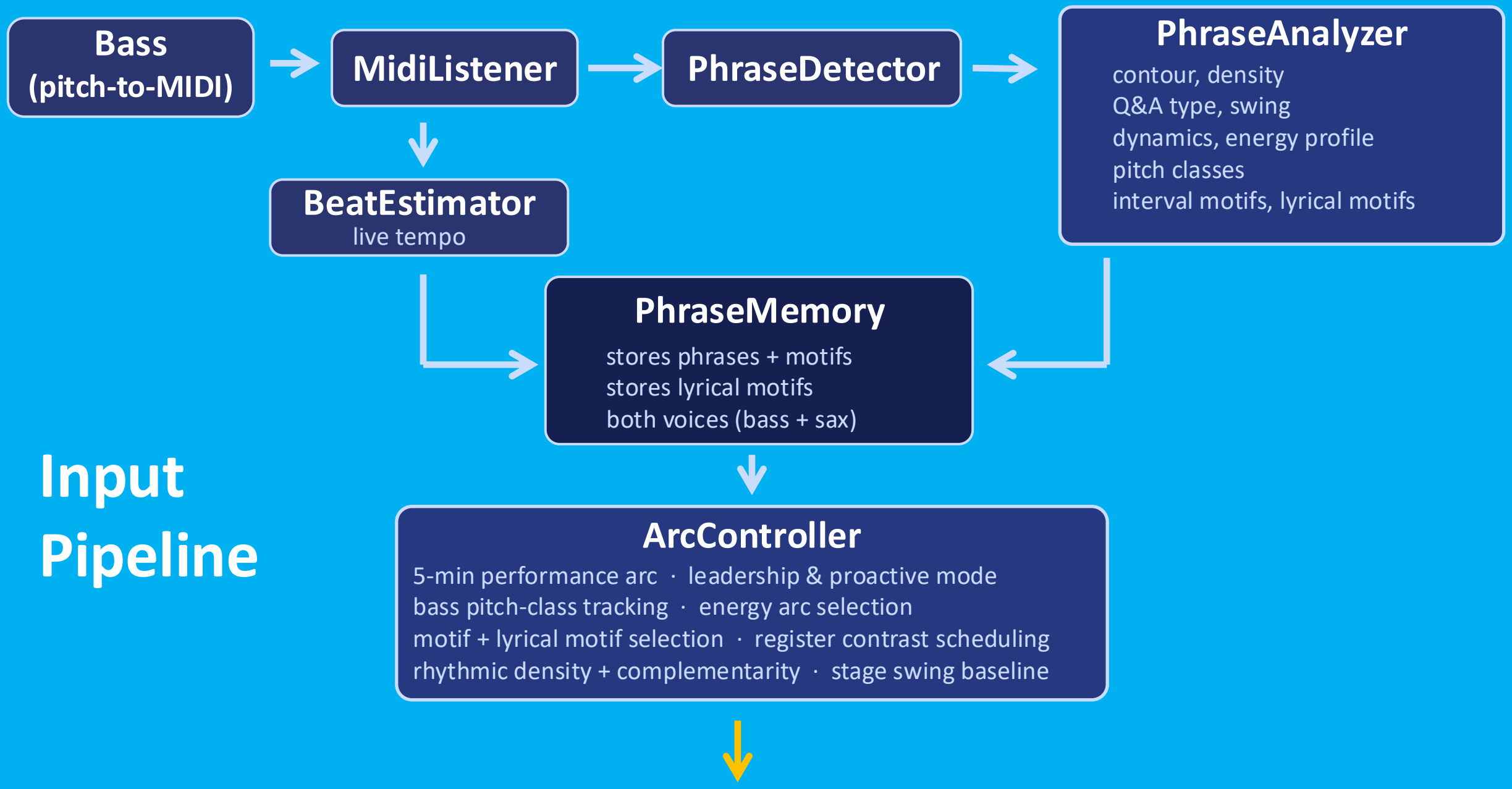
** Good name for a song?*

5 minute arc

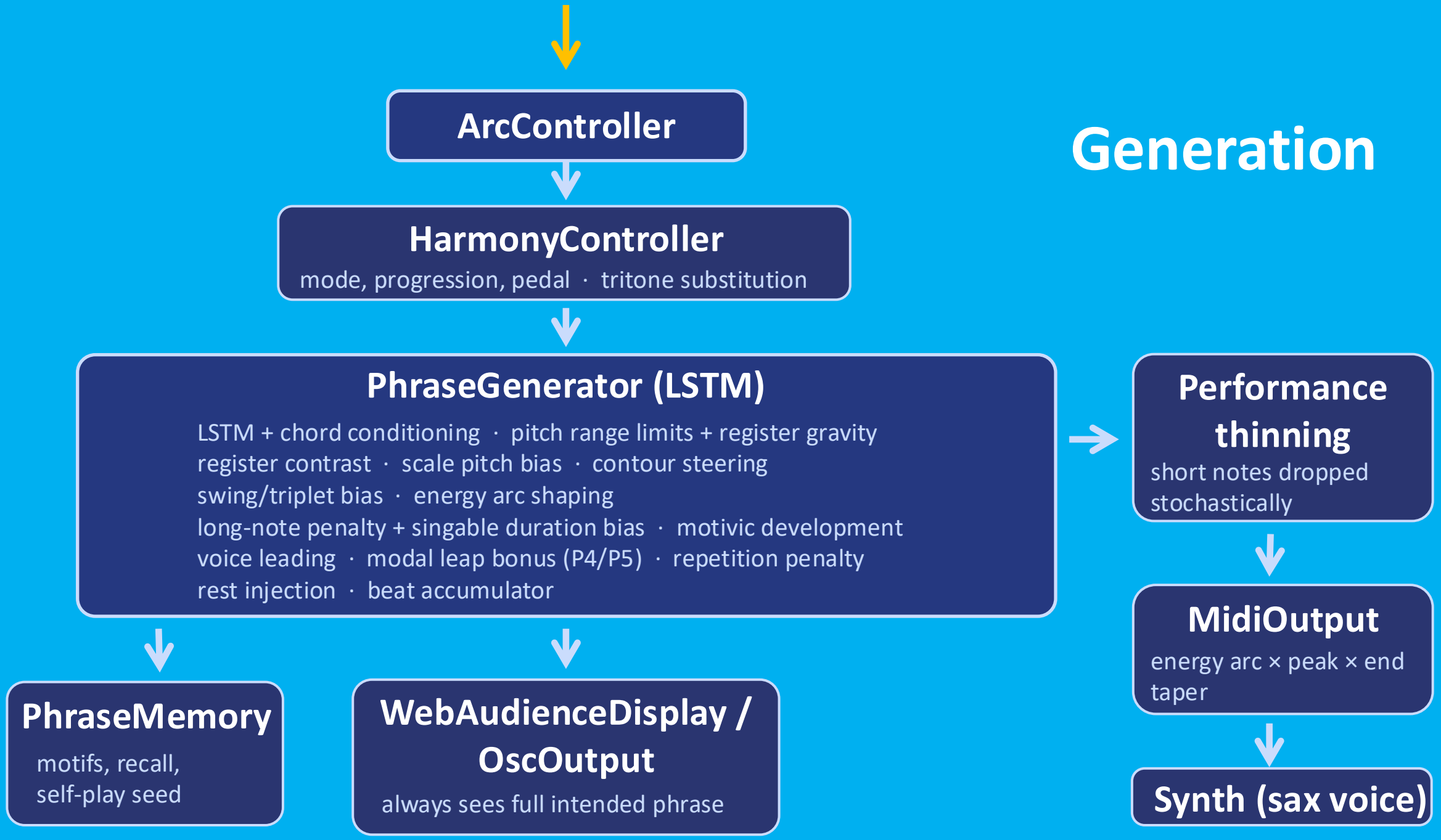
1. Sparse
   - Open, sustained, establish motifs slowly
2. Building
   - Mixed, lines begin to flow
3. Peak
   - Fast, active, max rhythmic intensity
4. Recapitulation
   - Lyrical return, recall themes with space
5. Resolution
   - Slowest, long tones, fading out

wolfson.numbersintonotes.net

Input Pipeline



Generation


ArcController
HarmonyController
mode, progression, pedal · tritone substitution
PhraseGenerator (LSTM)
LSTM + chord conditioning · pitch range limits + register gravity
register contrast · scale pitch bias · contour steering
swing/triplet bias · energy arc shaping
long-note penalty + singable duration bias · motivic development
voice leading · modal leap bonus (P4/P5) · repetition penalty
rest injection · beat accumulator
Performance thinning
short notes dropped stochastically
MidiOutput
energy arc × peak × end taper
Synth (sax voice)
PhraseMemory
motifs, recall, self-play seed
WebAudienceDisplay / OscOutput
always sees full intended phrase

- Repo is on github.com/davidderoure/wolfson

- Acknowledgements:
  - Ray d'Inverno (for jazz, Lisp, AI, slick licks)
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  - Bofan Ma (for always enabling musicality)
  - Wolfson Digital Research Cluster (for AIMC 2024)

- *Grand Challenges for Computing Research* – Tony Hoare

Potential limitations of the chosen approach

- Time feel might be mechanical – lack of microtiming
- Conversation might be too polite
- No personal voice
- Arc is clock-driven, not music-driven
- Motivic development might feel like repetition
- It uses MIDI, so perhaps it inevitably sounds like a very good MIDI file but there'll be a lack of expression/articulation